

MODEL DEVELOPMENT

COVID-19, Turkey
from April 1, 2020 to December 31, 2020

FA51 - Prof. Robbins
March 13, 2026



MINERVA[®]
UNIVERSITY

Variables

Independent Variable

- Time (t), in days

Dependent Variables

- $S(t)$ (Susceptible)
- $I(t)$ (Infected)
- $R(t)$ (Removed) $\Rightarrow$ death+ recovery

Variable Type

- continuous variable

Initial Values

- $t_0 = 0$ day (April 1, 2020)
- N (total population of Turkey in 2020)= 83,614,362 people (Türkiye İstatistik Kurumu [TÜİK], 2020)
- I_0 (confirmed active cases. By April 1, 2020)= 15,679 people (T.C. Sağlık Bakanlığı, 2020)
- R_0 (represents the sum of deaths (277) and recoveries (333) reported as of April 1, 2020)= 610 people (T.C. Sağlık Bakanlığı, 2020)
- S_0 ($N - R_0 - I_0$) $\approx$ 83,583,713 people

Parameters

Transmission Rate (b)

time-dependent parameter (Bayram et al., 2023)

Period 1: April 1 – May 31 (Lockdown)

Period 2: June 1 – Sept 30 (Normalization & Tourism)

Period 3: Oct 1 – Dec 31 (Second Wave Restrictions)

Recovery Rate (k)

$k = 1/14$

(14-day standard quarantine period, (Demirbilek et al., 2020))

References

Bayram, G. E., Bayram, A. T., & Sak, M. (2023). Learn from the past, COVID-19 and beyond: The case of Türkiye. Journal of Mediterranean Tourism Research, 3(1), 37–48. <https://digitalcommons.usf.edu/cgi/viewcontent.cgi?article=1025&context=jometr>

Demirbilek, Y., Pehlivan Türk, G., Özgüler, Z. Ö., & Alp, M. E. (2020). COVID-19 outbreak control, example of ministry of health of Turkey. Turkish Journal of Medical Sciences, 50(SI-1), 489–494. <https://doi.org/10.3906/sag-2004-187>

Wikipedia contributors. (2026). Timeline of the COVID-19 pandemic in Turkey. Wikipedia. https://en.wikipedia.org/wiki/Timeline_of_the_COVID-19_pandemic_in_Turkey

Worldometer. (2020). Turkey COVID-19 statistics. <https://www.worldometers.info/coronavirus/country/turkey/>

Türkiye İstatistik Kurumu (TÜİK). (2020). Adrese dayalı nüfus kayıt sistemi sonuçları, 2020. <https://veriportali.tuik.gov.tr/Bulten/Index?p=AdreseDayali-Nufus-Kayit-Sistemi-Sonuclari-2020-37210>

T.C. Sağlık Bakanlığı. (2020). Genel koronavirüs tablosu. <https://covid19.saglik.gov.tr/tr-66935/genel-koronavirus-tablosu.html>